

Group Topic: Big Data Analysis for Rainfall-Based Flood Risk Prediction in Albay using SYNOP Data	Lab No. 2
Git Repo/ Colab Link: https://github.com/PotatoKevz/About-bda	Date: 04/24/26

Objectives

- Conceptualize the data gathering method
- Look for online sites that can be requested for SYNOP data
- Decode the downloaded raw data reports in a CSV format
- Clean the decoded data reports
- Visualize the cleaned data reports

Tasks Completed

1. Looked for sites that can be requested by SYNOP Dataset, and the group arrived at OGIMET (*Organización Meteorológica Iberoamericana*), an online platform that provides access to historical weather observations, especially SYNOP reports from meteorological stations worldwide.

The screenshot shows the OGIMET website, which is a professional information service about meteorological conditions. The page layout includes a header with the OGIMET logo and navigation links. The main content area is divided into sections for 'Weather Model Forecast', 'METEOROLOGICAL DATA ANALYSIS', 'UNDECODED REPORTS', and 'TEXT INFORMATION'. A sidebar on the right provides quick access to various data types and reports.

2. Scraped data from weather stations near Albay; Albay (Lezapi City), Camarines Norte (Daet), Camarines Sur (Naga), Catanduanes (Virac), Masbate, Romblon, and Quezon (Infanta and Tayabas) Stations:

	98427 Tayabas (Quezon) 75k.csv 	 jab2023-6181-30913	Apr 23	8.2 MB	
	98434 Infanta (Quezon) 75k.csv 	 jab2023-6181-30913	Apr 23	8.2 MB	
	98440 Camarines Norte 76k.csv 	 jab2023-6181-30913	Apr 23	8.5 MB	
	98442 Camarines Sur Station 11k.csv 	 jab2023-6181-30913	Apr 23	3.5 MB	
	98444 Albay Station 76k.csv 	 jab2023-6181-30913	Apr 23	8.4 MB	
	98446 Catanduanes 76k.csv 	 jab2023-6181-30913	Apr 23	8.2 MB	
	98536 Romblon 74k.csv 	 jab2023-6181-30913	Apr 23	7.9 MB	
	98543 Masbate 76k.csv 	 jab2023-6181-30913	Apr 23	8 MB	

Scraped Datasets from OGIMET

	A	B	C	D	E	F	G
1	WMOIND	YEAR	MONTH	DAY	HOUR	MIN	REPORT
2	98444	2000	1	1	0	0	AAXX 01001 98444
3	98444	2000	1	1	3	0	AAXX 01031 98444
4	98444	2000	1	1	6	0	AAXX 01061 98444
5	98444	2000	1	1	9	0	AAXX 01091 98444
6	98444	2000	1	1	12	0	AAXX 01121 98444
7	98444	2000	1	1	15	0	AAXX 01151 98444
8	98444	2000	1	1	18	0	AAXX 01181 98444
9	98444	2000	1	2	0	0	AAXX 02001 98444
10	98444	2000	1	2	3	0	AAXX 02031 98444
11	98444	2000	1	2	6	0	AAXX 02061 98444
12	98444	2000	1	2	9	0	AAXX 02091 98444
13	98444	2000	1	2	12	0	AAXX 02121 98444
14	98444	2000	1	2	15	0	AAXX 02151 98444
15	98444	2000	1	2	18	0	AAXX 02181 98444
16	98444	2000	1	2	21	0	AAXX 02211 98444
17	98444	2000	1	3	0	0	AAXX 03001 98444
18	98444	2000	1	3	3	0	AAXX 03031 98444
19	98444	2000	1	3	6	0	AAXX 03061 98444
20	98444	2000	1	3	9	0	AAXX 03091 98444
21	98444	2000	1	3	12	0	AAXX 03121 98444
22	98444	2000	1	4	0	0	AAXX 04001 98444
23	98444	2000	1	4	3	0	AAXX 04031 98444
24	98444	2000	1	4	6	0	AAXX 04061 98444
25	98444	2000	1	4	9	0	AAXX 04091 98444
26	98444	2000	1	4	12	0	AAXX 04121 98444
27	98444	2000	1	4	15	0	AAXX 04151 98444

CSV Content of Albay Station (Raw SYNOT Data Report)

```

C:\Users\romeo\Desktop\Big Data Analysis\Programs\Flood-prediction> ogimet_synop_scraper.py -c _chunks
56 BASE_URL = "http://www.ogimet.com/cgi-bin/getsynop"
57 COLUMNS = ["WMOIND", "YEAR", "MONTH", "DAY", "HOUR", "MIN", "REPORT"]
58
59 logging.basicConfig(
60     level=logging.INFO,
61     format="%(asctime)s [%(levelname)s] %(message)s",
62     datefmt="%Y-%m-%d %H:%M:%S",
63 )
64 log = logging.getLogger(__name__)
65
66 # Mimic a real browser - OGIMET rejects bot-like user agents
67 HTTP_HEADERS = {
68     "User-Agent": (
69         "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
70         "AppleWebKit/537.36 (KHTML, like Gecko) "
71         "Chrome/124.0.0.0 Safari/537.36"
72     ),
73     "Accept": "text/plain, text/html, */*",
74     "Accept-Language": "en-US,en;q=0.9",
75     "Connection": "keep-alive",
76 }
77
78
79 # --- helpers ---
80
81 def fmt(dt: datetime) -> str:
82     return dt.strftime("%Y%m%d%H%M")
83
84
85 def _chunks(start_year: int, end_year: int, days: int) -> Iterator[Tuple[datetime, datetime]]:
86     """Yield (begin, end) datetime pairs in "days"-sized steps."""
87     now = datetime.now(timezone.utc).replace(tzinfo=None)

```

Code Snippet of Data Scraper

3. Cleaned the decoded SYNOP datasets:

 CleanedAlbay.csv 	 jab2023-6181-30913	6:37 PM	5.4 MB	
 CleanedCamNorte.csv 	 jab2023-6181-30913	6:42 PM	5.4 MB	
 CleanedCatanduanes.csv 	 jab2023-6181-30913	6:42 PM	5.3 MB	
 CleanedInfanta.csv 	 jab2023-6181-30913	6:43 PM	5.3 MB	
 CleanedMasbate.csv 	 jab2023-6181-30913	6:43 PM	5.4 MB	
 CleanedRomblon.csv 	 jab2023-6181-30913	6:43 PM	5.2 MB	
 CleanedTayabas.csv 	 jab2023-6181-30913	6:44 PM	5.3 MB	

Cleaned Datasets

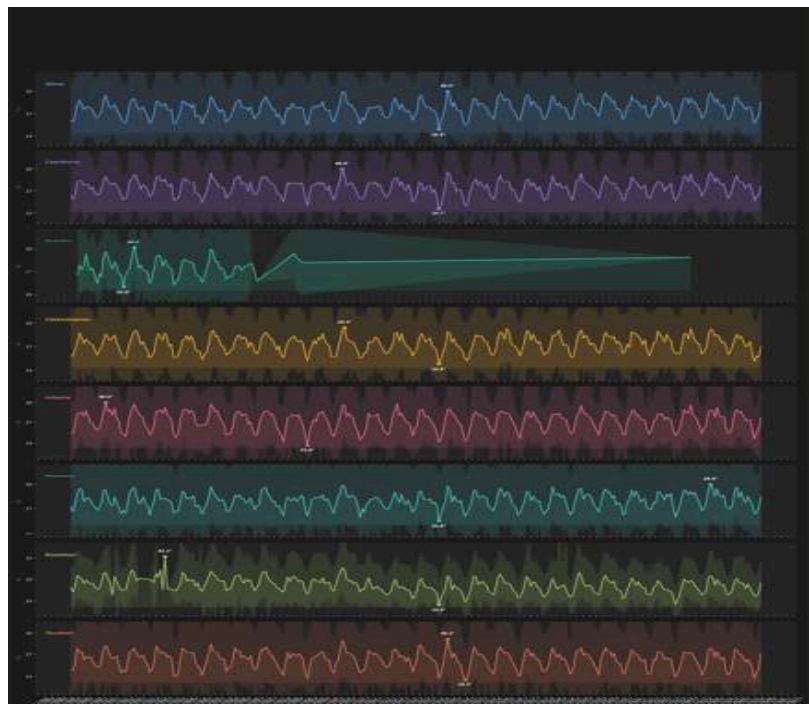
```

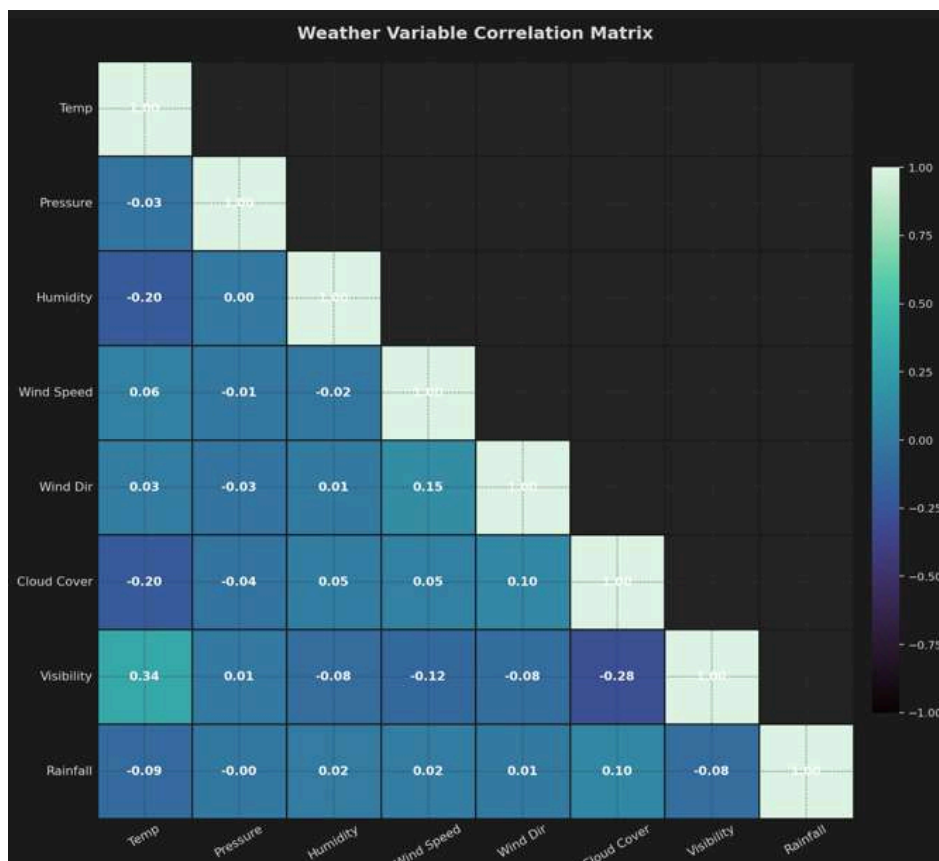
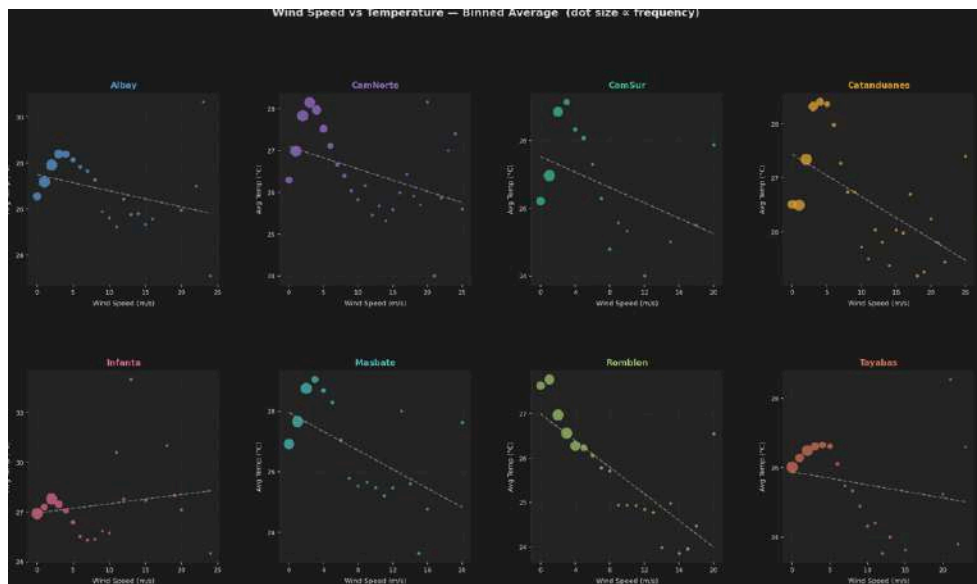
C: > Users > romeo > Desktop > Big Data Analysis > Programs > flood-prediction > synop_decoder.py > ...
62
63 def decode_rix(group: str) -> dict:
64     """Section 0 indicator group iRixhVW (5 chars).
65
66     iR (precipitation indicator):
67         0/1/2 group 6 present somewhere → no sentinel (use decoded value)
68         3 precip amount = 0, no group 6 → rain_sentinel = 0.0 (Case B)
69         4 station does not measure precip → rain_sentinel = None (Case A)
70     """
71     out = {}
72     if len(group) != 5:
73         return out
74     ir = group[0]
75     if ir == "3":
76         out["rain_sentinel"] = 0.0 # "no rain reported" → 0.0 mm
77     elif ir == "4":
78         out["rain_sentinel"] = None # truly missing (station doesn't measure)
79     # ix - station type / present-weather indicator (ignored)
80     # h - height of cloud base (code table 1600)
81     h_code = group[2]
82     h_table = {"0": 0, "1": 50, "2": 100, "3": 200, "4": 300,
83               "5": 600, "6": 1000, "7": 1500, "8": 2000, "9": 2500, "/": None}
84     out["cloud_base_m"] = h_table.get(h_code)
85     # VV - visibility (code table 4377)
86     vv = group[3:5]
87     out["visibility_m"] = _decode_vv(vv)
88     return out
89
90
91 def _decode_vv(vv: str) -> float | None:
92     """Convert W code (00-99) to metres (WMO Code Table 4377)."""

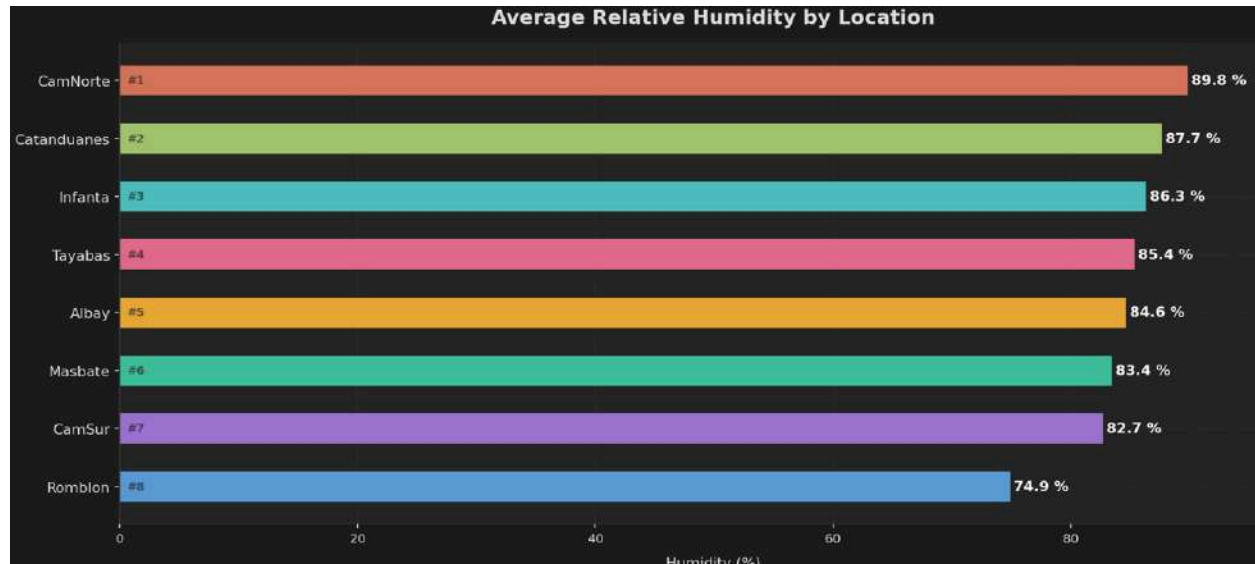
```

Code Snippet

4. Visualization of the datasets:







Results and Analysis

Summary of Findings

- **Thermal Stability:** All eight stations maintain a consistent tropical range between 26.3°C (**TAYABAS**) and 28.0°C (**MASBATE**), with a clear seasonal peak in May - June (~28.7°C).
- **Regional Anomalies:** January 2014 was identified as a significant regional cold event, showing record lows across five different stations simultaneously.
- **Humidity & Geography:** Humidity levels are high across the mainland (82% - 90%) but **ROMBLON** acts as a significant outlier **74.9%**, likely due to its island geography and rain-shadow effects.
- **Wind Patterns:** Mean daily wind speeds peak during the Northeast Monsoon (Amihan) in December and January. The data accurately reflected extreme weather, such as the **60.0m/s** gust recorded in Camarines Sur in September 2000.

Challenges and Solutions

Challenge: Difficulty in finding and gathering a large enough dataset (e.g., 500k+ records) for big data analysis.

Solution: Developed a Python-based data scraper to automatically collect historical records from the OGIMET platform , allowing

Challenge: Complex SYNOP Encoding. Raw SYNOP data is highly compressed and unreadable for standard analysis tools.

Solution: A custom Python decoder was developed to parse specific code tables (e.g., WMO Code Table 4377 for visibility) into human-readable metric units.

Challenge: Missing Rainfall Indicators. Different stations use varied sentinel values for reporting "no rain" versus "data missing".

Solution: The decoding logic was programmed to differentiate between these states—assigning 0.0 mm for reported no rain and none for truly missing measurements to ensure data integrity.

Conclusion

This week's tasks successfully moved the project from the planning stage to active data management. By establishing an automated pipeline to scrape and decode SYNOP data, the group has secured the necessary high-fidelity information required for rainfall-based flood prediction. The objectives for data gathering, decoding, and initial visualization were fully met